INFORMATION SECURITY LAB · MINI PROJECT

USB Forensics Investigation Report

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Case	#1 - Investigation of scenario_b.img
Examiner	Vijay
Case opened (UTC)	2026-10-29 20:38:20
Report generated (UTC)	2026-10-29 20:38:21
Evidence source	/home/claude/src/usb_forensics_toolkit/data/scenario_b.img
Tool	usb_forensics_toolkit writeblock.py v1.0 (software write-block)

1. Key findings

Evidence integrity	VERIFIED · re-verified after analysis: PASS
Was the evidence device genuine?	GENUINE - SanDisk Cruzer Blade serial 4C530001230915101234 matches its known-good fingerprint in all 2 session(s)
Rogue / spoofed devices	**LIKELY_SPOOFED** session #4 0951:1666 "Kingston DataTraveler 3.0" at 2026-10-08 22:44:10 (score 130)
What was copied, and when	**Session #2** 2026-10-08 10:15:32 - 10:42:10: notes.txt, holiday.jpg **Session #5** 2026-10-08 22:50:02 - 23:10:10 **AFTER HOURS**: client_list.docx, salary.xlsx (deleted), passwords.txt (deleted), report.pdf, invoice.jpg, secret.txt
Files	8 found · 2 deleted (2 recovered) · 6 flagged
Alerts	9 (7 HIGH, 2 MEDIUM, 0 LOW)

Indicators are triage aids for the examiner; they must be interpreted with the wider case context and corroborated before conclusions are drawn.

2. Evidence acquisition and integrity

Status	**VERIFIED**
Acquired (UTC)	2026-10-29 20:38:20
Working copy	/home/claude/src/usb_forensics_toolkit/evidence/case_1/evidence.img
Size	67,108,864 bytes
SHA-256 source (before)	`849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b`
SHA-256 source (after)	`849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b`
SHA-256 working copy	`849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b`
MD5 source / copy	`1332d927c87a7233f1609b6abd5b32d4` `1332d927c87a7233f1609b6abd5b32d4`
Write protection	software: O_RDONLY, write probe BLOCKED (EBADF: descriptor opened O_RDONLY); OS: running as root: file permissions do NOT stop writes; hardware write-blocker: not used / not verified

3. Chain of custody

Time (UTC)	Examiner	Action	Details
2026-10-29 20:38:20	Vijay	ACQUISITION_STARTED	`source=/home/claude/src/usb_forensics_toolkit/data/scenario_b.img dest=/home/claude/src/usb_forensics_toolkit/evidence/case_1/evidence.img tool=usb_forensics_toolkit writeblock.py v1.0 (software write-block)`
2026-10-29 20:38:20	Vijay	SOURCE_HASHED	`sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b md5=1332d927c87a7233f1609b6abd5b32d4 size=67108864`
2026-10-29 20:38:20	Vijay	WRITE_PROTECTION	`software: O_RDONLY, write probe BLOCKED (EBADF: descriptor opened O_RDONLY); OS: running as root: file permissions do NOT stop writes; hardware write-blocker: not used / not verified`

Time (UTC)	Examiner	Action	Details
2026-10-29 20:38:20	Vijay	ACQUISITION_VERIFIED	`copy=/home/claude/src/usb_forensics_toolkit/evidence/case_1/evidence.img sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b md5=1332d927c87a7233f1609b6abd5b32d4 (copy set to read-only 0444)`
2026-10-29 20:38:20	Vijay	ANALYSIS_STARTED	`usb_forensics_toolkit 2.0; pytsk3 on the verified copy /home/claude/src/usb_forensics_toolkit/evidence/case_1/evidence.img; usb_log=/home/claude/src/usb_forensics_toolkit/data/usb.log; pcap=/home/claude/src/usb_forensics_toolkit/data/usb_capture.pcap`
2026-10-29 20:38:21	Vijay	REVERIFY_PASS	`sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b md5=1332d927c87a7233f1609b6abd5b32d4 stored_sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b`
2026-10-29 20:38:21	Vijay	ANALYSIS_COMPLETED	`files=8 deleted=2 flagged=6 sessions=5 read_errors=0`

4. Device identification and impersonation check

4.1 Imaged volume (boot sector)

File system	FAT32
Volume serial	4C53-B0B0
Label	USB_B
OEM name	mkfs.fat
Sector / cluster	512 / 512 bytes
Size	67,108,864 bytes

4.2 USB devices observed

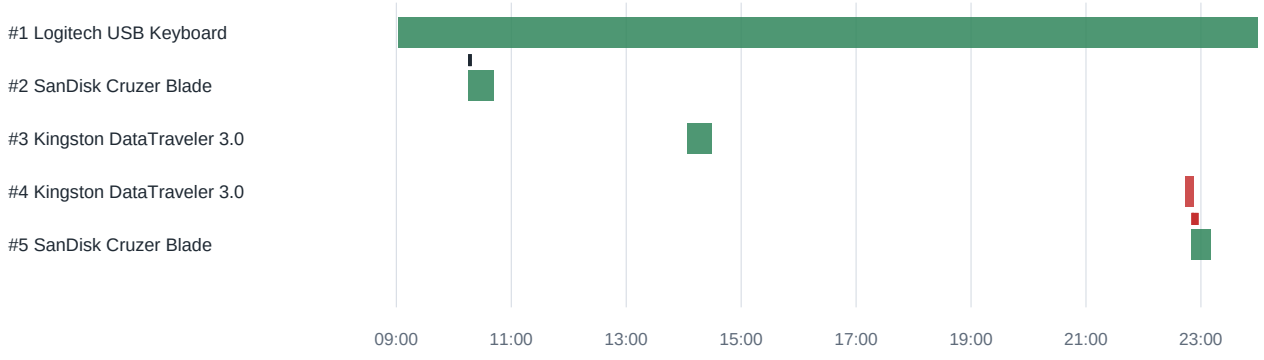
#	VID:PID	Device (claimed)	Serial	Interfaces	Verdict	Score
1	046d:c31c	Logitech USB Keyboard	`(none)`	hid_keyboard	**GENUINE**	0
2	0781:5567	SanDisk Cruzer Blade *imaged device*	`4C530001230915101234`	mass_storage	**GENUINE**	0
3	0951:1666	Kingston DataTraveler 3.0	`001CC0EC34A1F3B0`	mass_storage	**GENUINE**	0
4	0951:1666	Kingston DataTraveler 3.0	`(none)`	hid_keyboard	**LIKELY_SPOOFED**	130
5	0781:5567	SanDisk Cruzer Blade *imaged device*	`4C530001230915101234`	mass_storage	**GENUINE**	0

Session #4 0951:1666 - LIKELY_SPOOFED (score 130) · sources: kernel_log,usbmon

(+60) claims to be a storage device but enumerated a HID keyboard interface (BadUSB keystroke-injection pattern)
(+20) expected mass-storage interface was never bound by the kernel
(+25) no serial number (iSerialNumber=0); the genuine model always reports one
(+15) reports USB 1.10 but genuine Kingston DataTraveler 3.0 is USB 2.00+ (descriptor bcdUSB)
(+10) keyboard connected for only 48 s (typical of automated keystroke injection)

Scoring compares each device's descriptors (VID/PID, strings, serial, bcdDevice, bcdUSB, bound interfaces) with known-good fingerprints. GENUINE < 25 ≤ SUSPICIOUS < 60 ≤ LIKELY_SPOOFED. A perfect descriptor clone cannot be detected from descriptors alone.

5. Connection history



Bars: USB sessions coloured by verdict (green genuine, amber suspicious, red likely spoofed, grey unverified). Ticks: files written to the imaged device (red = flagged). Times are the suspect machine's local time.

#	Device	Port / disk	Connected	Disconnected	Notes
1	046d:c31c USB Keyboard	1-4	2026-10-08 09:02:11	-	still connected at end of log
2	0781:5567 Cruzer Blade	1-1 sdb	2026-10-08 10:15:32	2026-10-08 10:42:10	64.0 MiB, matches image
3	0951:1666 DataTraveler 3.0	1-1 sdc	2026-10-08 14:03:55	2026-10-08 14:30:00	14776.0 MiB
4	0951:1666 DataTraveler 3.0	1-3	2026-10-08 22:44:10	2026-10-08 22:44:58	**after hours**
5	0781:5567 Cruzer Blade	1-2 sdb	2026-10-08 22:50:02	2026-10-08 23:10:10	**after hours**, 64.0 MiB, matches image

6. File activity

Path	Size	Created	Modified	Session	State
/notes.txt	19	2026-10-08 10:16:05	2026-10-02 09:12:40	#2	-
/holiday.jpg	26	2026-10-08 10:17:41	2026-09-20 18:04:22	#2	-
/client_list.docx	26	2026-10-08 22:51:14	2026-10-06 16:20:08	#5	flagged
/salary.xlsx	26	2026-10-08 22:51:20	2026-10-05 17:30:12	#5	deleted, flagged
/passwords.txt	23	2026-10-08 22:52:03	2026-09-29 11:45:30	#5	deleted, flagged
/report.pdf	184	2026-10-08 22:53:40	2026-10-07 12:02:56	#5	flagged
/invoice.jpg	26	2026-10-08 22:55:12	2026-10-08 22:44:36	#5	flagged
/secret.txt	13	2026-10-08 22:56:31	2026-10-08 22:56:30	#5	hidden, flagged

On FAT, copying a file keeps its original *modified* time but sets a new *created* time, so created > modified indicates the content was copied onto the device rather than authored there. FAT stores the access date only and no deletion time.

6.1 Deleted and recovered files

Path	Recovered to	SHA-256 of content
/salary.xlsx	`/home/claude/src/usb_forensics_toolkit/recovered/case_1/14_salary.xlsx`	`b3a15aa00a8e34b5abac8e110c98d3cfdba6ca070e5f177e553945a72b03c905`
/passwords.txt	`/home/claude/src/usb_forensics_toolkit/recovered/case_1/11_passwords.txt`	`8212e599f490f7736bdff4e3d6af838c9b5eb91e06168dd24da8fabfb4d04ccd`

6.2 Suspicious files

Path	Reasons	SHA-256
/client_list.docx	suspicious keyword in filename	`f97e604274bfaa436eba17d35b10012feca6312c6500f247097e499de6c4772d`
/salary.xlsx	suspicious keyword in filename	`b3a15aa00a8e34b5abac8e110c98d3cfdba6ca070e5f177e553945a72b03c905`
/passwords.txt	suspicious keyword in filename	`8212e599f490f7736bdff4e3d6af838c9b5eb91e061688dd24da8fabfb4d04ccd`
/report.pdf	extension .pdf does not match zip/office content; suspicious keyword in content	`c0ab6c19601d62d00f11a2aed15ad642a9992bddb180f293a64750e64678a84d`
/invoice.jpg	extension .jpg does not match exe content	`c710cb27a101a3c65f91b2095818959373d9dd3a2be87473505481092b95c2be`
/secret.txt	hidden attribute set; suspicious keyword in filename	`011eb8d35cbda2fec2e9fcbfe794ae8fee5f368af4da129cd326aac5010e7179`

7. Alerts

Severity	Category	Verdict	Details
HIGH	impersonation	LIKELY_SPOOFED	session #4 0951:1666 Kingston DataTraveler 3.0 at 2026-10-08T22:44:10: score 130; claims to be a storage device but enumerated a HID keyboard interface (BadUSB keystroke-injection pattern); expected mass-storage interface was never bound by the kernel; no serial number (iSerialNumber=0); the genuine model always reports one; reports USB 1.10 but genuine Kingston DataTraveler 3.0 is USB 2.00+ (descriptor bcdUSB); keyboard connected for only 48 s (typical of automated keystroke injection)
HIGH	file	FLAGGED_COPY	/client_list.docx written to device serial 4C530001230915101234 at 2026-10-08T22:51:14 (after hours): suspicious keyword in filename

Severity	Category	Verdict	Details
HIGH	file	FLAGGED_COPY	/salary.xlsx written to device serial 4C530001230915101234 at 2026-10-08T22:51:20 (after hours); later deleted: suspicious keyword in filename
HIGH	file	FLAGGED_COPY	/passwords.txt written to device serial 4C530001230915101234 at 2026-10-08T22:52:03 (after hours); later deleted: suspicious keyword in filename
HIGH	file	FLAGGED_COPY	/report.pdf written to device serial 4C530001230915101234 at 2026-10-08T22:53:40 (after hours): extension .pdf does not match zip/office content; suspicious keyword in content
HIGH	file	FLAGGED_COPY	/invoice.jpg written to device serial 4C530001230915101234 at 2026-10-08T22:55:12 (after hours): extension .jpg does not match exe content
HIGH	file	FLAGGED_COPY	/secret.txt written to device serial 4C530001230915101234 at 2026-10-08T22:56:31 (after hours): hidden attribute set; suspicious keyword in filename
MEDIUM	policy	AFTER_HOURS	session #4 0951:1666 Kingston DataTraveler 3.0 connected outside office hours at 2026-10-08T22:44:10
MEDIUM	policy	AFTER_HOURS	session #5 0781:5567 SanDisk Cruzer Blade connected outside office hours at 2026-10-08T22:50:02

8. Reconstructed timeline

File-system and USB events in the suspect machine's local time; chain-of-custody actions in UTC (examiner clock).

Time	Source	Event	Detail
2026-09-20 18:04:22	filesystem	FILE_MODIFIED	/holiday.jpg (content timestamp, carried over from the source)
2026-09-29 11:45:30	filesystem	FILE_MODIFIED	/passwords.txt (content timestamp, carried over from the source)
2026-09-29 11:45:30	filesystem	FILE_DELETED_ENTRY	/passwords.txt (deletion time unknown, mtime used)
2026-10-02 09:12:40	filesystem	FILE_MODIFIED	/notes.txt (content timestamp, carried over from the source)
2026-10-05 17:30:12	filesystem	FILE_MODIFIED	/salary.xlsx (content timestamp, carried over from the source)
2026-10-05 17:30:12	filesystem	FILE_DELETED_ENTRY	/salary.xlsx (deletion time unknown, mtime used)
2026-10-06 16:20:08	filesystem	FILE_MODIFIED	/client_list.docx (content timestamp, carried over from the source)
2026-10-07 12:02:56	filesystem	FILE_MODIFIED	/report.pdf (content timestamp, carried over from the source)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/client_list.docx (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/holiday.jpg (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/invoice.jpg (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/notes.txt (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/passwords.txt (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/report.pdf (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/salary.xlsx (FAT stores access date only)
2026-10-08 00:00:00	filesystem	FILE_ACCESSED	/secret.txt (FAT stores access date only)
2026-10-08 09:02:11	usb_log	USB_CONNECT	#1 046d:c31c Logitech USB Keyboard serial=None [GENUINE]
2026-10-08 10:15:32	usb_log	USB_CONNECT	#2 0781:5567 SanDisk Cruzer Blade serial=4C530001230915101234 [GENUINE]
2026-10-08 10:16:05	filesystem	FILE_CREATED	/notes.txt -> written during USB session #2 (Cruzer Blade)
2026-10-08 10:17:41	filesystem	FILE_CREATED	/holiday.jpg -> written during USB session #2 (Cruzer Blade)
2026-10-08 10:42:10	usb_log	USB_DISCONNECT	#2 0781:5567 SanDisk Cruzer Blade serial=4C530001230915101234 [GENUINE]
2026-10-08 14:03:55	usb_log	USB_CONNECT	#3 0951:1666 Kingston DataTraveler 3.0 serial=001CC0EC34A1F3B0 [GENUINE]
2026-10-08 14:30:00	usb_log	USB_DISCONNECT	#3 0951:1666 Kingston DataTraveler 3.0 serial=001CC0EC34A1F3B0 [GENUINE]
2026-10-08 22:44:10	usb_log	USB_CONNECT	#4 0951:1666 Kingston DataTraveler 3.0 serial=None [LIKELY_SPOOFED] [AFTER HOURS]
2026-10-08 22:44:36	filesystem	FILE_MODIFIED	/invoice.jpg (content timestamp, carried over from the source)
2026-10-08 22:44:58	usb_log	USB_DISCONNECT	#4 0951:1666 Kingston DataTraveler 3.0 serial=None [LIKELY_SPOOFED] [AFTER HOURS]
2026-10-08 22:50:02	usb_log	USB_CONNECT	#5 0781:5567 SanDisk Cruzer Blade serial=4C530001230915101234 [GENUINE] [AFTER HOURS]
2026-10-08 22:51:14	filesystem	FILE_CREATED	/client_list.docx [SUSPICIOUS] -> written during USB session #5 (Cruzer Blade)
2026-10-08 22:51:20	filesystem	FILE_CREATED	/salary.xlsx [SUSPICIOUS] -> written during USB session #5 (Cruzer Blade)
2026-10-08 22:52:03	filesystem	FILE_CREATED	/passwords.txt [SUSPICIOUS] -> written during USB session #5 (Cruzer Blade)
2026-10-08 22:53:40	filesystem	FILE_CREATED	/report.pdf [SUSPICIOUS] -> written during USB session #5 (Cruzer Blade)
2026-10-08 22:55:12	filesystem	FILE_CREATED	/invoice.jpg [SUSPICIOUS] -> written during USB session #5 (Cruzer Blade)
2026-10-08 22:56:30	filesystem	FILE_MODIFIED	/secret.txt
2026-10-08 22:56:31	filesystem	FILE_CREATED	/secret.txt [SUSPICIOUS] -> written during USB session #5 (Cruzer Blade)
2026-10-08 23:10:10	usb_log	USB_DISCONNECT	#5 0781:5567 SanDisk Cruzer Blade serial=4C530001230915101234 [GENUINE] [AFTER HOURS]
2026-10-29 20:38:20	custody(UTC)	ACQUISITION_STARTED	Vijay: source=/home/claude/src/usb_forensics_toolkit/data/scenario_b.img dest=/home/claude/src/usb_forensics_toolkit/evidence/c
2026-10-29 20:38:20	custody(UTC)	SOURCE_HASHED	Vijay: sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b md5=1332d927c87a7233f1609b6abd5b32d4 size=671088

Time	Source	Event	Detail
2026-10-29 20:38:20	custody(UTC)	WRITE_PROTECTION	Vijay: software: O_RDONLY, write probe BLOCKED (EBADF: descriptor opened O_RDONLY); OS: running as root: file permissions do NO
2026-10-29 20:38:20	custody(UTC)	ACQUISITION_VERIFIED	Vijay: copy=/home/claude/src/usb_forensics_toolkit/evidence/case_1/evidence.img sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6
2026-10-29 20:38:20	custody(UTC)	ANALYSIS_STARTED	Vijay: usb_forensics_toolkit 2.0; pytsk3 on the verified copy /home/claude/src/usb_forensics_toolkit/evidence/case_1/evidence.i
2026-10-29 20:38:21	custody(UTC)	REVERIFY_PASS	Vijay: sha256=849bba6ebffdd19392c53d18d9232ccd2daa09b6dab45470f30a0ed43120669b md5=1332d927c87a7233f1609b6abd5b32d4 stored_sha2
2026-10-29 20:38:21	custody(UTC)	ANALYSIS_COMPLETED	Vijay: files=8 deleted=2 flagged=6 sessions=5 read_errors=0

9. Limitations

• Write-blocking is software-based (O_RDONLY descriptor, verified write probe; optional kernel read-only flag for live devices). No hardware write-blocker was used. Hashes detect change, they do not prevent it.

• FAT has no deletion timestamp and only an access date; the deletion entries in the timeline use the modified time as an approximation.

• Log and file-system times are local, zone-less wall-clock times. Clock drift on the suspect machine affects both equally but is not corrected.

• Impersonation checks rely on descriptors. A device that clones every descriptor of a genuine model (including its serial and interfaces) cannot be distinguished this way.

• Suspicious-file rules (signature mismatch, entropy, keywords, hidden attribute) can produce false positives and need manual review.

Examiner	Signature	Date
Vijay		